L10.2: Tangents for scalar-valued multivariable functions

Transcript Language

English

Hello, and welcome to the Maths 2 component of the Online BSc program on Data Science

and Programming. In this video, we are going to talk about tangents for scalar-valued multivariable

functions. So, we have seen the notion of tangents for functions of one variable. This

is a topic that we studied extensively when we read one variable calculus, and we used

it to study the ideas of the tangent line, and the idea of differentiability and the derivative. So, let us recall first what are tangent lines to curves.

So, a tangent line to a curve C at a point P on the curve C is a line which represents

the instantaneous direction in which the curve C moves at the point P. So, later on, we summarize

this information in terms of the derivative. And traditionally, we have thought of, we

think of tangents as lines, which just touch the curve at that point.

As we now know, the idea of just touches is a little vague, which is why you have to use

more involved definitions like the derivative. And as examples, here are curves that we had

actually seen in our study of one variable calculus, along with the tangents to those

curves. So, this is just a recollection of what we had done in one variable calculus.

So, from here, how did we move to functions?

So, we looked at the graph of the function. So, suppose we have f from D to R, which is

a function, where D is, of course, now a subset of R. So, if we assume that the graph is a curve and we take a point in this domain D then the tangent line to f at that point,

x is the tangent to the graph of that function f at the point x comma f of x, this was our

definition.

And just as an example, here is the function f of x is x cube and you can see these tangent

lines. So, if your function is nice, by which we mean it is differentiable then you can

compute the explicit equations for these tangent lines by looking at y minus f of a is equal

to f prime of a times x minus a, this is the equation of the tangent line at the point

a.

And we saw that this notion of just touches, can have interesting implications. For example,

for y is x cube, if you will draw the tangent at 0 then it is actually the x axis and that

actually cuts the curve of the, that actually cuts the graph of the function f of x is x

cube. So just touches can have various implications for what it means.

It can be like any of these lines, the red lines which are drawn or it can actually be

a line, which cuts the curve, and those are kind of special or interesting points, as

we later saw in our study of maxima and minimum. So, this is just a collection of tangents

for functions of one variable.

Now, we want to study tangents for scalar-valued multivariable functions. So, what do we mean

by tangents? What do, what does the collection of tangents look like and so on? So, we already

have some idea about this, when we studied the notion of partial derivatives and directional

derivatives because what we did there was we restricted ourselves to a particular line,

then we said, let us look at how the function looks on top of that line. And over there,

we said, well, now we, over here we ask what is the rate of change of the function when

we restricted to that line, which made sense because we had restricted to a function of

one variable. We will use the same idea here.

So, suppose you have for now let us start with two variables. So, suppose you have a

function of two variables, which is defined on a domain D in R2 and contains some open

ball around the point a comma b, which sometimes are represented by a tilde. So, consider a

line L in D passing through a tilde and restrict f to L. This is exactly what we did in when

we defined the notion of the directional derivative.

And since it is now a function of one variable, we can consider the tangent to f at a tiled

over L, as in the previous slide. By previous slide, I mean, since it is a function of one

variable we know exactly what we mean by tangent. It is the line which represents the instantaneous

direction of the graph. And that is the line that we consider.

So, if the line L is in the direction of the unit vector u, then the tangent if it exists

and we know examples from one variable calculus where it does not exist, will be the line

with slope f u of a tilde and passing through the point a tilde comma f of a tilde. So, let us understand this statement better. So, we know from one variable calculus, as we

just observed that the equation of the tangent is given by the derivative at that point,

so once we know the derivative at that point in one variable calculus that is, then we

can compute what is the equation of the tangent line.

So, now what we have done here is we have restricted ourselves to the line L. So, we

have a one variable function and then using that one variable function, what is the derivative,

that is exactly the directional derivative, where how do you compute the directional derivative,

well you take this line L, and you look at unit vector, which is pointing in the same

direction as that line L.

Of course, this line L may not pass to the origin. So, what you have to do is you have

to translate this line L parallelly back to the origin, and then on that line L, you have

to take a unit vector. Once you take this unit vector, you have to compute the directional derivative. And once we compute that directional derivative, you can use that to compute the

equations of the point, so it will be a line.

So, if this is the line L. And earlier when we had the one variable situation, this was

your x axis, this was your y axis, and so you had this line below, and the derivative

was something like this, so it was with respect to the x axis, when we took the derivative,

it told you the slope, meaning the angle with respect to the x axis.

So now your line is some arbitrary line in our R2, so what you do is above this line

you take a line with the slope prescribed by f u of a tilde. And so, this is a line

with that slope, passing through the point a tilde comma f of a tilde so this uniquely

determines the line. And let us see a couple of examples of this in GeoGebra.

So, let us look at the function, f of x, y is x, y. So, here is the function f of x,

y is x, y, so rather interesting looking function. Let us look at the point, let us say one comma one. So, if you look at the point one comma one. Let us take a plane passing through that

point, according to this unit vector. So, I have some unit vector, which I am going

to change now as this changes.

So, here is a plane passing through the point 1 comma 1. So, I hope you can see the plane.

And now let us restrict the graph of this function to that plane. So, if we do that,

then we get this curve. So, let me let me remove the, so I hope you can see the curve

on the plane. So let me remove the graph. So, here is the graph, which is 1 and here

is how the how it looks like on that plane. And now if I remove that plane, you will see

further that this is how it looks like on the line below that, which is passing through

the point 1 comma 1.

And now, we can ask what is the tangent to this add 1 comma 1. So, if we compute what

that is, well, here is how that tangled looks like. So, here is the tangent. So, I hope you can see the tangent. And we can vary this as the unit vector u varies, which is the

same as saying as the plane vary.

So, I will bring back the plane. So, this is all happening on this plane. It is clear

I hope from the picture. And as I vary my plane or my unit vector, rather, you will

see how this changes. So, as the plane changes the corresponding, so these are all planes

passing through that same point as the unit vector changes, the corresponding part of

the graph changes that you have changes and so the tangent changes. So, this is how it

looks like.

Let us do another example. So instead of these, suppose now I have the graph of the function,

sine of x, y. So, this was again a function that we have seen before. So, here is sine

of x, y. Let us look at the point maybe 1 comma 0.

So, here is a plane passing the point 1 comma 0. I hope you can see it. And now let us see what the intersection of these two is that gives you the following curve, and I will

remove this graph so you can see the curve. So, we have seen this curve before, as well.

So, this is the curve passing through 1 comma 0.

And now we can ask what is the tangent to that. So, if you look at the tangent, here

is how the tangent looks like. So, this is at the point 1 comma 0, and I can vary my

unit vector. And as I vary my unit vector, the plane changes accordingly, the part of

the graph which is intersected with the plane changes, and accordingly the tangent line

changes. So, I hope you can see what is happening. So, this is just a visual demonstration of

what we said in that previous slide.

So, the main point here is that, if you have this line L, you get the corresponding plane

above that, and on that plane you have this line, which has slope f u of a tilde, and

which passes to the point a tilde comma f of a tilde. So, using these facts we can write down the equation of this line.

And let us try to write down what that equation is. So, we can write this equation down in

several ways. So, first of all, what is the line below? So, I am going to take u to be

a u1 comma u2. So, this is the unit vector on that line. So, unit vector on L and then

a, b is my point. So, a, b so a tilde equal a, b is the point where I want to compute

the equation of the tangent line. So, fu a, b is the directional derivative at that point.

So, these are the, these are the, this is everything that I know. So first of all, what

is the equation of the line? So, for the line, so first of all, we know that it is on the

x, y plane, so that means z is zero, this is one of the defining factors. And the other

is that it passes through the point a comma b, and it is in the direction of the unit

vector, u1 comma u2.

So, since that is the case, we can write down the equation as so u1 times y minus b is equal to u2 times x minus a. And how do I check that this is indeed correct? Well, you can

see that the slope is u2 by u1 provided of course that you u1 is non-zero, and if u1

is 0 you can see that it is the correct line which is x is equal to a.

And if you plug-in the values of x and y to a and b respectively then this equation is

satisfied because you have 0 is equal to 0 that means it passes through a comma b. So,

it is the intersection of these two planes. So z0 is a plane that is x, y plane, and u1

times y minus b is u2 times x minus a is another plane, so it passes through that plane.

And in fact, if I remove this restriction that z is 0, then that is a plane above the

line L. So, the corresponding plane which I will denote by P is u1 times y minus b is

u2 times x minus a. So now, this is what I know. So, instead of writing L like this,

I can write it as a parametric equation and this is often very useful, so that we know

what is happening.

So, the other way of writing L is as xt is equal to a plus t times u1, yt is equal to

b plus t times u2, and zt is equal to 0 because it is on the x, y plane, so this is how you

write L. And how did I get this?

Well, this is if you remember from linear algebra, lines are meaning any arbitrary line

is an affine flat, and how do I get those by translating some line which passes through

the origin, which means a subspace of dimension one. And with subspace is that? That is the

line which is passing through the vector u1 comma u2. So, based on that what I can say

is that xt, yt, zt is exactly just to reiterate this, a, b 0 plus t times u1, u2, 0.

So, this is saying that if you have you are shifting the line passing through u1 comma

u2 comma 0, because it is on the x, y axis, x, y plane, so that it passes through a comma

b. This was exactly the idea of affine flats. And we are going to use the same idea now

to write down the equation of the tangent line as well.

So, what happens for the tangent line. So, we have identified the line L and the plane

P. So, the tangent line is actually on this plane P, and it is at an angle of a fu meaning

the tangent of the angle, the tan of the angle, which it makes with L is fu of a comma b.

So, parsing that out, what it means is, if on that line you move a unit distance, which

means that you move by that unit vector then the z coordinate will move by fu a comma b.

So, what that means is, just to put that in perspective, what that means is that the parametric

equations will be given by xt is a plus tu1, yt is d plus tu2. So, this you expected because

when you project that line down the tangent line down you will get this line L, and we

have written down the parametric equations for that. So, the question is what happens

to zt.

And zt, as we just saw, if you move a unit distance, which means if t is one so that

means you're moved by that unit vector then the z coordinate will move by fu of a comma

b. So, then I can write this, as f of a comma b plus t times fu of a comma b, so these are

the parametric equations for the tangent line.

Now you can rewrite this in various ways. For example, you can rewrite this in the following

form x minus a by u1 is y minus b by u2 is z minus f of a comma b by fu of a comma b.

Of course, you assume that these are non-zero, if they are 0 there is a way of interpreting

this, etc. So, this is called the symmetric equations for the line. Not very important,

but I am just mentioning it. And the other way of writing it is the way we have viewed

this as an affine flat and that is really the very interesting.

So, what we are saying is that, this line is a line passing through a comma b comma

f of a comma b, so xt, yt, zt just to rewrite this is a comma b comma f of a, b. So, this

is a point plus t times u1 comma u2 comma fu of a comma b. And what is this saying?

This is exactly corresponding to the equation here.

So, again this is, so this line is in affine flat, it passes through a comma b comma f

of a comma b that is why I have translated it by this point. And if you translate it

back to the origin, what line do you get, you get the line passing through u1 comma

u2 comma f of fu of a comma b. Why is that? Because the line here was u1 comma u2 comma

0, it was a line passing through u1 comma u2 comma 0.

And the line that you have your makes an angle of theta where tan theta is fu of a comma

b. So, if you translate that you get this same line here and that is what it say that

it is the line passing through this vector here, that is exactly what we mean by the

derivative. So, this explains why we get this line. So again, this is the vector form of

you can often called the vector form.

So, these are the equations of the tangent line. So, I hope this explanation shed some

light on how we obtain these tangent line equations. So, let us see a couple of examples so that we can make this concrete.

So, let us compute some of these things in the at the mentioned points and in the mentioned

directions. So, let us look at the function x plus y and the tangent at 1 comma 1 in the

direction of 1 comma 0. So, how do I get this? So, for this, I have to first find the directional

derivative.

So, let us find the directional derivative. So, the directional derivative, so first I

should find del f del x, which is actually what this is at 1 comma 1. So, add x comma

y del f del x is just 1. So, her and then, so what I want here is, so this is exactly

fu because u is 1 comma 0, so this is fu at any point so I am particular at 1 comma 1.

So fu at 1 comma 1 is 1 and now we can write down the equations.

So, maybe, let me do it in the vector form that is the easiest. So, in the vector form

it is xt, yt, zt is a comma b comma f of a, b, so f of a, b is

1 because a and b are 1 respectively. So, f of 1 comma 1 is two, plus t times u1 comma

u2 which his up here, comma fu of a comma b, fu of a comma b we have computed as 1,

so this is what we get. So, the other way of writing this is to say that xt is 1 plus

t, yt is 1, and zt is 2 plus t.

So, and now you can try and visualize this in GeoGebra. I will encourage you to do that.

Let us do the next one, which is f of x, y is xy. We actually saw this example a few

minutes ago. So here, we have to compute this in the direction of 3 comma 4, so first, we

need a unit vector in that direction.

So, what is a unit vector in that direction? Well, you can compute it is 3 by 5 comma 4

by 5, then we need to compute what is fu of 1 comma 1. So, to do that let us use since

it is a nice function, I can use my gradient formula. So, what is the gradient of f at

x comma y it is so differentiating with respect to x, this is y with respect to y it is x,

so gradient f at 1 comma 1 is just 1 comma 1 again.

And then fu of a comma b is so fu of 1 comma 1 is one times u1, so one-times 3 by 5, plus

1 times 4 by 5, which is 7 by 5. So, now I can write down my equations. So, in the parametric

form or in the vector form I get xt, yt, zt is the point, so 1 comma 1 comma f of the

point, so which is 1 again, because f of 1 comma 1 is 1 plus t times u1 comma u2, 3 by

5 comma 4 by 5 comma fu of 1 comma 1, which is 7 comma 5, 7 by 5.

So, which gives us 1 plus 3t by 5 comma 1 plus 4t by 5 comma 1 plus 7t by 5. And from

here, you can get the parametric form. So, I hope you can see it is not at all difficult

to compute these once we know how to do a directional-derivatives, fine. The final example

is you have sine of x, y. I think, we have computed this before.

So, in the direction of 1 comma 2. So first what is a unit vector? So, this is u is 1

over root 5, 1 comma 2. And then what is the gradient? So, the gradient of f at x, y is

y times cosine x, y comma x times cosine x, y. So, if we compute this at pi comma 1, we get a gradient of f at pi comma 1 is cosine of pi, which is minus 1 and then pi times

cosine of pi, which is minus pi.

So, fu of at pi comma 1 is minus 1 times a 1 by root 5, plus minus pi times 2 by root

5, which is some fairly nasty looking expressions. So, minus 2 pi minus 1 by root 5. So, this

is fu of pi comma 1, and now we can easily write down what is the vector form of this

tangent line.

So, xt, yt, zt is you take your points, so pi comma 1 comma evaluate the function at

this point, sine of pi, sine of pi is 0 plus t times u1, sorry u1 is 1 by root 5 not 1

comma 2 by root 5, comma minus 2 pi minus 1 by root 5, and now you can get the parametric

form as well. So, I hope these examples are somewhat illustrative of how to compute these

tangent lines.

So, let us now go ahead and do the same thing for general functions of n variables. So, we did this in two variables because it is easy to visualize this in terms of pictures,

but now we can do this for n variables.

Now, we cannot really visualize this any longer, but the ideas are exactly the same. So, suppose

you have a function f of x tilde, which is defined on a domain D in Rn containing some

open ball around your point a tilde you take a line restrictive f to that line passing

a tilde that is, and since it is a function of one variable, we can consider the tangent

to f at a tilde over L as before.

So, if the line L is in the direction of the unit vector u, then the tangent if it exists

will be the line with slope fu of a tilde and passing through the point a tilde comma

f of a tilde. And from here we can find out the equation of the line.

So, how do we do that? So, this is very similar to the two-variable case. And what we can

do is now suppose, I have n variables. So, the n plus 1th variable, let me denote by xn plus 1. So now, I have or let me denote the n plus 1 variable by z. So, the first

n plus 1 variables are x1, x2, xn and the n plus 1th variable, which we are using to

measure the function, let us call that z. So, z is the variable in which we are measuring

the function.

So, then, exactly the same way as we have done for the previous case, we can look at

the line passing through the point a tilde. So, a tilde is a1, a2, an, and which is in

the direction of the unit vector u. So, u is u1, u2, un. So, the line through a tilde

in the direction of u is for the parametric equations are x of t, so xi of t is ai plus

t times ui. So again, a line passing through u1, u2, un it will be t times u1, u2, un and

now you are translating it by the vector a.

So, that is for i that this happens. So, in terms of the vector form. And of course, I

have to also say what happens for the coordinates z, so z is 0. So, in other words, in the coordinate form this is x tilde of t comma z of t is a tilde comma 0 plus t times u comma 0.

So, z is not playing any role, because everything is right now it is still the line in the x1,

x2, xn space. So, now, this line, I want to take the line which is at an angle of theta,

where theta is tan of theta is fu of a tilde. And once again, what that means is, so therefore,

the tangent line to f at a tilde above L, so if I call this above line L, so this this

line I am calling L.

So, above L is, so the first n coordinates remain the same and then for z we know how

much it jumps. So, if x the first n coordinates move by a unit vector then z moves by fu,

this is exactly what we mean by rate of change. So, that means what we will get is it x-tilde

of t comma z of t is equal to a tilde comma f of a tilde plus t times u comma fu of a

tilde.

And now you can write down the parametric form. So, this is the vector form. So, for the parametric form you will have xi of t is ai plus t times ui, exactly as we had for

the line L and the remaining coded is a t, so for zt we will have f of a tilde plus t

times fu of a tilde. So, we can write it in either which way, the vector form or the parametric

form.

Let us do an example. So, this might make it easier to view. So, what you do remember

is that these are the two forms for the tangent line. So, either this which is the vector

form or the second one, which is the parametric form. So, you have f of x, y is, so you have

f of x, y, z is x, y, plus y, z plus z, x and let us see what is the tangent at 1,1,1

in the direction of minus 1 minus 2,2. So first, we will compute the unit vector. So,

if you compute the unit vector that is one-third times minus 1 comma minus 2 comma 2.

So now, we need to know what is the directional derivative in this direction, at the point

1 comma 1 comma 1. So, for that we need the gradient. So, let us compute what is the gradient

of x, y, z. So, the gradient of x, y, z is y plus z, z plus x, x plus y. So, at 1 comma

1 comma 1. This is 2 comma 2 comma 2. So, from here, we can compute the vector equation

so that is going to be xt, yt, zt. I am using zt here, already.

So, the function, let us view it in terms of u, so that is a last variable. So, ut,

this is the point which is 1 comma 1 comma 1 comma the function evaluated at 1 comma

1 comma 1, which is 3 plus t times the unit vector 1. So minus one-third, minus two-third

comma fu. And what is fu? So, I have just computed the gradient. So, what is fu? So,

fu at 1 comma 1 comma 1 is gradient dot with u, so that is 2 comma 2 comma 2 dot with one-third

minus 1 comma minus 2 comma 2, so which is minus two-third.

So, here I will get minus two-thirds. So, the vector equation will be xt is 1 minus

t by 3, yt 1 minus 2t by 3, zt is 1 plus 2t by 3 and ut which is what represents the function

that is 3 plus or 3 minus 2t by 3. So, this is very easy. Once we know what to compute, it is very easy to actually compute it. Of course, we have implicitly used here that

we know that the gradient is continuous, which is why we can compute the directional derivative

fairly easily.

So, let us now add a word of caution, because from here, it may seem that all is hunky dory.

So, let us remember that tangents need not always exist. So, we have seen this even in

one dimension. So, if you take f of x, y is x, y by x squared plus y squared, if x, y

is not 0, 0, and 0 of x, y is 0, 0 we know that at the point 0, 0 there is some problem.

So, we have seen in this example that the partial derivatives exist, so gradient f at

0, 0 well, for u1, comma u2 is 0 if it is one of the unit vectors in the x or y direction

is e1 plus minus e1 or plus minus e2, but it does not exist otherwise.

And my bad, this is f sorry, I meant to say fu of at 0, 0. And so, once we know that the

partial derivative does not. The directional derivative in the particular direction does not exist, we know from our one variable calculus that the derivative of a one variable function

exists is precisely the same as saying that the tangent line exists otherwise, there is

no notion of a tangent line.

There is no line, which captures the behavior of the instantaneous direction. So, that means

the tangent lines in all directions other than

along the x or y axis at 0, 0 do not exist, that is exactly what it means. Let us look

at the function mod x plus mod y, this is maybe even crazier. And here, it is not going

to exist for many, many directions.

So, for most directions or let us say at for many directions at many points, the tangent

link will not exist. I will suggest you do the algebra yourself, but instead, we look

at a picture of this graph and I think you will appreciate what I mean.

So, here is a picture of the graph of this function. This is the function mod x plus

mod y, and you can see it has lots and lots of corners, it has faces, and then it has corners. So, if you are along a face then then maybe you have a chance that a tangent

does exist because that will be like some kind of a constant function, but if you move

your plane even a little that will be a lot of trouble because then you will have lots

of corners and then the tangent may not exist. So just to give you an idea of what I am saying

here let us intersect this with some plane.

So, if I look at the corresponding graph, you can see how it looks like. I will remove

the graph of the function, here is how it looks like. You can see there is lots of edges,

and it is not even at some point it is even becoming, it is not even touching each other

and there are corners at various places. And as we move the plane, you will see that the

graph has lots of problems.

So, let us switch this animation on, and you can see, it looks very much like the mod x

is mod y. See, it always has some jagged part. So, if you are at those points you will have trouble. So, I hope this animation convinces you that there is a lot of places where the

tangent and there are a lot of directions that in which the tangent will not exist.

So, we can ask when do all the tangents exist? And so, this is equivalent to asking when

do all the directional derivatives exist? And this is a question that we actually know

the answer for. So, if f of x1, x2, xn is a function defined in our domain D in Rn containing

some open ball around the point a tilde, then we have this wonderful theorem, that if the

gradient exists and is continuous on some open ball around the point a tilde then for

every unit vector u, the directional derivative fu of a tilde exists and equals gradient f

of f at a tilde dot with u. And this says that the directional derivative exists, and

we have a formula to compute it, and that exactly means that the tangent exists in that

direction, and that we can write down its equation, which we have done a few slides

ago.

So, conclusion is that all the tangents at a point a tilde exist when gradient f exists

and is continuous on some open ball around the point a tilde. And once that happens,

we know how to write down the equation of the tangent line. Thank you.